# Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

## Answer 1 of 8 · Question 1 part e · 2 marks

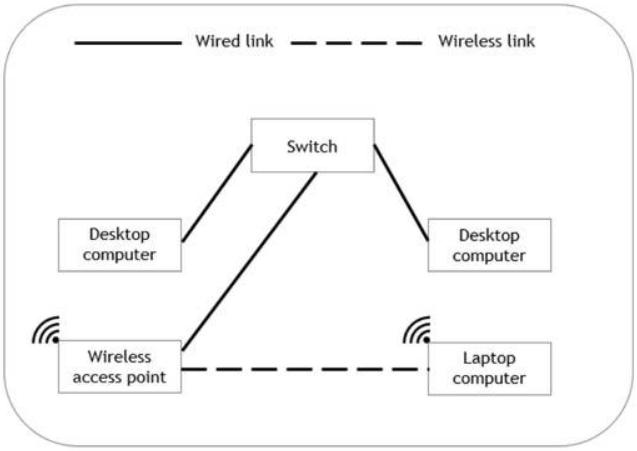
2025 JUN · Paper 01 · extract 1 of 4 · find part e in this mark scheme extract

<table>
<tr><td>1(a)</td><td>Award one mark for each correct row:<br><br><table><tr><th>Statement</th><th>DNS</th><th>PAN</th><th>WAN</th><th>WWW</th></tr><tr><td>A network of networks not limited to a single location</td><td></td><td></td><td>✓</td><td></td></tr><tr><td>A network that connects devices within the range of a single user</td><td></td><td>✓</td><td></td><td></td></tr><tr><td>A set of hypertext documents accessed through the HTTP protocol</td><td></td><td></td><td></td><td>✓</td></tr><tr><td>A system that maps domain names to IP addresses</td><td>✓</td><td></td><td></td><td></td></tr></table></td><td>Do not award a mark for a row containing more than one tick.</td><td>4</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(b)</td><td>Award one mark for each of:<br><br>• Transport (layer) (1)<br>• (Data) link / network link / network interface (layer) (1)</td><td>Accept<br>• transfer or transmission for Transport<br><br>Do not accept:<br>• Network layer and Application layer (in the question)<br>• Data access layer, Internet layer (alternative names for network layer)<br>• Data’ or ‘network’ without ‘link’ or ‘interface’</td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(c)</td><td>The only correct answer is D<br><br>A is not correct because HTTP is used to transfer hypertext<br>B is not correct because it is used to view email held on a server<br>C is not correct because it is used to download email from a server</td><td></td><td>1</td></tr>
</table>

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 1 of 8 · Question 1 part e · 2 marks

2025 JUN · Paper 01 · extract 2 of 4 · find part e in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(d) | Award **one** mark for each correct connection:<br>• One solid line from wireless access point to switch (1)<br>• One solid line from each desktop to switch (1)<br>• One dashed line from laptop computer to wireless access point (1)<br>  | No marks should be awarded if any additional lines have been drawn for each mark point.<br>Do not double penalise (e.g. if you have penalised a second line from desktop to WAP then don't penalise extra line from WAP to desktop) | 3 |

# Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

## Answer 1 of 8 · Question 1 part e · 2 marks

2025 JUN · Paper 01 · extract 3 of 4 · find part e in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e) | Award up to **two** marks for a linked description such as:<br>• To find weaknesses in a system (1) fix/repair weakness (1)<br>• To find vulnerabilities in a system (1) by testing the effectiveness of the security measures in place (1)<br>• An ethical hacking attack on a network (1) to identify vulnerabilities / recommend measures/take action to improve security (1)<br>• Tests whether users of the network (1) can be tricked into giving away information / giving criminals access to the network / downloading malware (1)<br>• To test the system security (1) / sanctioned by the owner (1) | Explanation can be a combination of different marking points. E.g. MP1a and MP2b | 2 |
| | | **Total for question 1** | **12** |

# Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

## Answer 1 of 8 · Question 1 part e · 2 marks

2025 JUN · Paper 01 · extract 4 of 4 · find part e in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 2 of 8 · Question 5 part c · 2 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| 5(a) | <table><tr><th>P</th><th>Q</th><th>R</th><th>Q OR R</th><th>P AND (Q OR R)</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table><br>Award **one** mark for each of:<br>• All possible combinations in columns P, Q and R (1)<br>• Correct values in Q or R (1)<br>• Correct values in final column (1) | Allow follow through for incorrect mark point 2. | 3 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(b) | B AND S AND NOT L<br>Award **one** mark for each of:<br>NOT L or B AND S (1)<br>Fully correct (1) | • Any order and ignore brackets | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(c) | Any **two** of:<br>• Remove code vulnerabilities in programming languages (1).<br>• Eliminate bad programming practices (1). | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(d) | Award **one** mark each for any of:<br>• Copyright (1)<br>• Patent (1)<br>• Licensing (1)<br>• Trademark (1) | | 3 |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 2 of 8 · Question 5 part c · 2 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

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Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 2 of 8 · Question 5 part c · 2 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 3 of 8 · Question 2 part d.i · 1 mark

2021 NOV · Paper 01 · extract 1 of 4 · find part d.i in this mark scheme extract

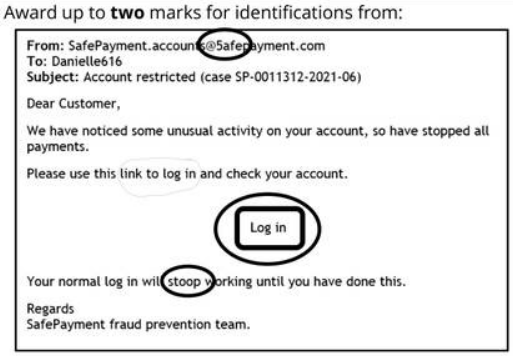
| | | | |
|---|---|---|---|
| 2(a) | **The only correct answer is B**<br><br>*A is not correct because a password is needed to prevent unauthorised access to the network*<br>*C is not correct because it only provides security after access to the network*<br>*D is not correct because it only provides security after access to the network* | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(b)(i) | Award **one** mark from:<br><br>• the email should be addressed to Danielle by her (full) name (1)<br>• the email should be personalised (1) | | 1 |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 3 of 8 · Question 2 part d.i · 1 mark

2021 NOV · Paper 01 · extract 2 of 4 · find part d.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(b)(ii) | Award up to **two** marks for identifications from:<br>  <br>**From:** SafePayment.account@5afepayment.com<br>**To:** Danielle616<br>**Subject:** Account restricted (case SP-0011312-2021-06)<br>Dear Customer,<br>We have noticed some unusual activity on your account, so have stopped all payments.<br>Please use this link to log in and check your account.<br>Log in<br>Your normal log in will stoop working until you have done this.<br>Regards<br>SafePayment fraud prevention team. | Allow link to log in OR circle around button | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c)(i) | Award up to **two** marks for a linked description such as:<br><br>A hacker/third party spies on/watches the user (of an electronic device) (1) In order to obtain their personal identification number/password/login information/sensitive information (1) | | 2 |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 3 of 8 · Question 2 part d.i · 1 mark

2021 NOV · Paper 01 · extract 3 of 4 · find part d.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 2(c)(ii) | Award up to **two** marks for a linked explanation such as:<br><br>• tilt the screen away from possible viewers/position yourself with your back to a wall (1) to ensure no one can see the screen (1)<br>• shield your screen/keypad/keyboard when entering (sensitive/personal) information (1) to stop people seeing/memorising passwords/named sensitive item/sensitive/personal information (1)<br>• use long/strong passwords (1) to prevent onlookers memorising them as you type (1)<br>• use a screen/privacy filter (1) because it will prevent anyone not sitting directly in front of the screen from reading what is displayed (1)<br>• sit where the information displayed on the screen (1) can't be captured on CCTV/by a drone-mounted camera or viewed by someone using binoculars (1) | Allow 1 mark for don't enter private information in a public place | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(d)(i) | **The only correct answer is B**<br><br>*A is not correct because this is not the purpose of penetration testing*<br>*C is not correct because this is not the purpose of penetration testing*<br>*D is not correct because this is not the purpose of penetration testing* | | 1 |
| 2(d)(ii) | Award up to **two** marks for a linked description such as:<br><br>• compromised/unpatched software is more vulnerable to attack (1) and may allow an attacker control of the whole network (1)<br>• unpatched software has known weaknesses (1) which can be exploited by a hacker (1) | | 2 |
| | | **Total** | **11** |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 3 of 8 · Question 2 part d.i · 1 mark

2021 NOV · Paper 01 · extract 4 of 4 · find part d.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 4 of 8 · Question 6 part c.i · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part c.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(a)(i) | Award up to **two** marks for a linked explanation such as:<br>• Can transfer data quickly / reduced chance of packet collisions (1) as data only flows in one direction (1)<br>• No need for a server/switch/hub (1) because the packets do not have to be directly routed to a specific device / each workstation controls connectivity / packets are passed from workstation to workstation until the destination is reached (1)<br>• Every workstation gets equal access to resources (1) because each station has to wait until it gets a token / devices do not have to compete to get a token (1)<br>• Additional workstations can be easily added/easy to set up (1) because each workstation only connects to two other workstations (1)<br>• Easy to find faults (1) all of the tokens will end up on one workstation (1)<br>• Cheap to set up (1) uses minimum cabling (1) | Not a **Comparison** without expansion that fits mark scheme | 2 |
| 6(a)(ii) | Award up to **two** marks for a linked explanation such as:<br>• It needs to be easily scalable (1) mesh topology allows this as it is decentralised / nodes connect with other nodes around them (1)<br>• Can handle high volumes of data traffic (1), because data can travel via multiple routes (1)<br>• Is self-healing/resilient/allows alternative paths (1), which means data will still reach its destination even if a node or connection fails(1) Not system/computer/switch etc<br>• Enables it to span a huge geographic area (1), because additional nodes can be added to expand coverage (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(b)(i) | Award **one** mark for:<br>• Personal area network / PAN/WPAN | | 1 |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 4 of 8 · Question 6 part c.i · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part c.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(b)(ii) | Award up to **two** marks for a linked explanation such as:<br><br>• A faster connection speed (1) because fewer users/devices sharing the bandwidth/connection (1)<br>• Improved security /stated security issue(1) because it uses secure cellular data connection / not on public network /Santiago has to approve users(1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(c)(i) | Award **one** mark from:<br><br>• A record of activities/specified activity that have taken place on a computer system (1)<br>• Automatic record of what has happened and who did it (1) | | 1 |
| 6(c)(ii) | Award **one** mark from:<br><br>• To identify suspicious/malicious activity/changes (1)<br>• To increase accountability (1)<br>• To trace a problem back to its source/perpetrator (1)<br>• To find out if any users are using unauthorised applications (capable of putting the network at risk) (1) | | 1 |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 4 of 8 · Question 6 part c.i · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(d) | **Ethical hacking**<br>• Ethical hackers are white hat hackers<br>• Attempt to access the network as a hacker does<br>• Don't attempt to change or steal data<br>• Looking for weaknesses in the network<br>• Weakness pointed out<br>• Weaknesses fixed<br>• Could be employed by the business<br>• Could work for another specialist company<br>• Can include penetration testing<br><br>**Commercial analysis tools**<br>• Software used to find weaknesses<br>• Can be configured to check for a range of weaknesses<br>• Results/reports generated identifying faults<br>• Weaknesses fixed<br><br>**Review of network and user policies**<br>• Collection of rules and guidelines that govern the behaviours of network devices/users<br>• Need reviewing because may not comply with new laws and regulations<br>• Reviews should be scheduled | | 6 |
| | | **Total for question 6** | **15** |

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | No rewardable content. |
| Level 1 | 1–2 | Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion will contain basic information with little linkage between points made. |
| Level 2 | 3-4 | Demonstrates adequate knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion shows some linkages and lines of reasoning with some structure. |
| Level 3 | 5-6 | Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented.<br><br>The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured. |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 4 of 8 · Question 6 part c.i · 1 mark

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Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 5 of 8 · Question 6 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part c.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(a)(i) | Award up to **two** marks for a linked explanation such as:<br>• Can transfer data quickly / reduced chance of packet collisions (1) as data only flows in one direction (1)<br>• No need for a server/switch/hub (1) because the packets do not have to be directly routed to a specific device / each workstation controls connectivity / packets are passed from workstation to workstation until the destination is reached (1)<br>• Every workstation gets equal access to resources (1) because each station has to wait until it gets a token / devices do not have to compete to get a token (1)<br>• Additional workstations can be easily added/easy to set up (1) because each workstation only connects to two other workstations (1)<br>• Easy to find faults (1) all of the tokens will end up on one workstation (1)<br>• Cheap to set up (1) uses minimum cabling (1) | Not a **Comparison** without expansion that fits mark scheme | 2 |
| 6(a)(ii) | Award up to **two** marks for a linked explanation such as:<br>• It needs to be easily scalable (1) mesh topology allows this as it is decentralised / nodes connect with other nodes around them (1)<br>• Can handle high volumes of data traffic (1), because data can travel via multiple routes (1)<br>• Is self-healing/resilient/allows alternative paths (1), which means data will still reach its destination even if a node or connection fails(1) Not system/computer/switch etc<br>• Enables it to span a huge geographic area (1), because additional nodes can be added to expand coverage (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(b)(i) | Award **one** mark for:<br>• Personal area network / PAN/WPAN | | 1 |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 5 of 8 · Question 6 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part c.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(b)(ii) | Award up to **two** marks for a linked explanation such as:<br><br>• A faster connection speed (1) because fewer users/devices sharing the bandwidth/connection (1)<br>• Improved security /stated security issue(1) because it uses secure cellular data connection / not on public network /Santiago has to approve users(1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(c)(i) | Award **one** mark from:<br><br>• A record of activities/specified activity that have taken place on a computer system (1)<br>• Automatic record of what has happened and who did it (1) | | 1 |
| 6(c)(ii) | Award **one** mark from:<br><br>• To identify suspicious/malicious activity/changes (1)<br>• To increase accountability (1)<br>• To trace a problem back to its source/perpetrator (1)<br>• To find out if any users are using unauthorised applications (capable of putting the network at risk) (1) | | 1 |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 5 of 8 · Question 6 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(d) | **Ethical hacking**<br>• Ethical hackers are white hat hackers<br>• Attempt to access the network as a hacker does<br>• Don't attempt to change or steal data<br>• Looking for weaknesses in the network<br>• Weakness pointed out<br>• Weaknesses fixed<br>• Could be employed by the business<br>• Could work for another specialist company<br>• Can include penetration testing<br><br>**Commercial analysis tools**<br>• Software used to find weaknesses<br>• Can be configured to check for a range of weaknesses<br>• Results/reports generated identifying faults<br>• Weaknesses fixed<br><br>**Review of network and user policies**<br>• Collection of rules and guidelines that govern the behaviours of network devices/users<br>• Need reviewing because may not comply with new laws and regulations<br>• Reviews should be scheduled | | 6 |
| | | **Total for question 6** | **15** |

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | No rewardable content. |
| Level 1 | 1–2 | Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion will contain basic information with little linkage between points made. |
| Level 2 | 3-4 | Demonstrates adequate knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion shows some linkages and lines of reasoning with some structure. |
| Level 3 | 5-6 | Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented.<br><br>The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured. |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 5 of 8 · Question 6 part c.ii · 1 mark

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Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 6 of 8 · Question 6 part d · 6 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part d in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(a)(i) | Award up to **two** marks for a linked explanation such as:<br>• Can transfer data quickly / reduced chance of packet collisions (1) as data only flows in one direction (1)<br>• No need for a server/switch/hub (1) because the packets do not have to be directly routed to a specific device / each workstation controls connectivity / packets are passed from workstation to workstation until the destination is reached (1)<br>• Every workstation gets equal access to resources (1) because each station has to wait until it gets a token / devices do not have to compete to get a token (1)<br>• Additional workstations can be easily added/easy to set up (1) because each workstation only connects to two other workstations (1)<br>• Easy to find faults (1) all of the tokens will end up on one workstation (1)<br>• Cheap to set up (1) uses minimum cabling (1) | Not a **Comparison** without expansion that fits mark scheme | 2 |
| 6(a)(ii) | Award up to **two** marks for a linked explanation such as:<br>• It needs to be easily scalable (1) mesh topology allows this as it is decentralised / nodes connect with other nodes around them (1)<br>• Can handle high volumes of data traffic (1), because data can travel via multiple routes (1)<br>• Is self-healing/resilient/allows alternative paths (1), which means data will still reach its destination even if a node or connection fails(1) Not system/computer/switch etc<br>• Enables it to span a huge geographic area (1), because additional nodes can be added to expand coverage (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(b)(i) | Award **one** mark for:<br>• Personal area network / PAN/WPAN | | 1 |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 6 of 8 · Question 6 part d · 6 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part d in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(b)(ii) | Award up to **two** marks for a linked explanation such as:<br><br>• A faster connection speed (1) because fewer users/devices sharing the bandwidth/connection (1)<br>• Improved security /stated security issue(1) because it uses secure cellular data connection / not on public network /Santiago has to approve users(1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(c)(i) | Award **one** mark from:<br><br>• A record of activities/specified activity that have taken place on a computer system (1)<br>• Automatic record of what has happened and who did it (1) | | 1 |
| 6(c)(ii) | Award **one** mark from:<br><br>• To identify suspicious/malicious activity/changes (1)<br>• To increase accountability (1)<br>• To trace a problem back to its source/perpetrator (1)<br>• To find out if any users are using unauthorised applications (capable of putting the network at risk) (1) | | 1 |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 6 of 8 · Question 6 part d · 6 marks

2022 JUN · Paper 01 · extract 3 of 4 · find part d in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 6(d) | **Ethical hacking**<br>• Ethical hackers are white hat hackers<br>• Attempt to access the network as a hacker does<br>• Don't attempt to change or steal data<br>• Looking for weaknesses in the network<br>• Weakness pointed out<br>• Weaknesses fixed<br>• Could be employed by the business<br>• Could work for another specialist company<br>• Can include penetration testing<br><br>**Commercial analysis tools**<br>• Software used to find weaknesses<br>• Can be configured to check for a range of weaknesses<br>• Results/reports generated identifying faults<br>• Weaknesses fixed<br><br>**Review of network and user policies**<br>• Collection of rules and guidelines that govern the behaviours of network devices/users<br>• Need reviewing because may not comply with new laws and regulations<br>• Reviews should be scheduled | | 6 |
| | | **Total for question 6** | **15** |

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | No rewardable content. |
| Level 1 | 1–2 | Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science.<br><br>The discussion will contain basic information with little linkage between points made. |
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Answer 6 of 8 · Question 6 part d · 6 marks

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## Answer 7 of 8 · Question 1 part d · 1 mark

2023 JUN · Paper 01 · extract 1 of 4 · find part d in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(a) | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>1. Access (shared) peripherals (printers) (1)<br>2. Access the Internet (1)<br>3. Access shared file storage (server, cloud) (1)<br>4. Allows data/files/information to be shared (1)<br>5. Provide communication (email, social media, online meeting, chat) (1)<br>6. Provide collaboration (editing online documents, playing games) (1)<br>7. Allow hot-desking (1)<br>8. Allows centralised software updates/backups (1) | Award any example that can be categorised to a unique reason | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b)(i) | **The only correct answer is C**<br><br>***A*** *is not correct because compression does not encrypt a file*<br><br>***B*** *is not correct because* encryption does not ensure both sides use the same key<br><br>***D*** *is not correct because* encryption does not stop hackers accessing a network | | 1 |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 7 of 8 · Question 1 part d · 1 mark

2023 JUN · Paper 01 · extract 2 of 4 · find part d in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b)(ii) | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>1. (Electronic/combination) lock on doors (1)<br>2. Swipe/key cards/biometrics (1)<br>3. Security guards/security patrol (1)<br>4. CCTV cameras / surveillance technology (1)<br>5. Asset tagging/marking (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b)(iii) | Award **one** mark for any of the following:<br><br>1. It exploits/relies on human behaviours/tricks people (to reveal sensitive information) (1) | Do not award examples e.g. Pharming without further information. | 1 |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 7 of 8 · Question 1 part d · 1 mark

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| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(c) | **The only correct answer is D**<br><br>***A*** *is not correct because 5G does not have a* **high transmission latency**<br><br>***B*** *is not correct because a 5G does not have* ***built-in security***<br><br>***C*** *is not correct because* 5G does not have a **lower bandwidth than 3G and 4G** | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(d) | Award **one** mark for:<br><br>• Testing each block of code/subprogram/function independently (as it is written/finished) (1) | Do not accept module in isolation as this is in the question. | 1 |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 7 of 8 · Question 1 part d · 1 mark

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| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e) | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>1. Belong to/have membership in a professional society (1)<br>2. Attend computer science related conferences/gatherings (1)<br>3. Attaining/gaining training/educational opportunities (1)<br>4. Behave in ethical/legal/moral ways (1)<br>5. Stay up to date with changes in the field/read up-to-date publications (1)<br>6. Use responsible programming practices e.g. due diligence / testing / commenting code for maintainability (1)<br>7. Avoid bias when making design choices (1) | Award examples if they can be categorised to a unique bullet point<br><br>Answers must focus on computer science rather than being generic<br><br>Do not accept things that would be expected as part of an employee's role e.g. writing code | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 8 of 8 · Question 4 part a.iii · 2 marks

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| | | | |
|---|---|---|---|
| 4(a)(i) | Award **one** mark for any of the following up to a maximum of **two** marks:<br>• It has high bandwidth/transmission speed (1)<br>• It has low latency (1)<br>• It has low energy consumption (1)<br>• It has high capacity (1)<br>• It has high security (1) | Allow 'more users can be connected' for MP4 | **2** |
| Question | Answer | Additional Guidance | Mark |
| 4(a)(ii) | Award up to **two** marks for a linked description such as:<br>• Biometric identifiers (fingerprint, iris, face) (1) matched against those stored in the app/database (1)<br>• Two-factor authentication / 2FA (1) requiring users to enter a code delivered to a separate system (1)<br>• Users can be asked for memorable information/security question (1) that only they will know the answer to (1) | System in MP2b may be on a separate device or a separate app | **2** |
| Question | Answer | Additional Guidance | Mark |
| 4(a)(iii) | Award **one** mark for any of the following up to a maximum of **two** marks:<br>• To identify any bad programming practices/shortcuts/bugs (1)<br>• To highlight code vulnerabilities / enhance security (1)<br>• To make sure code is robust/will deal with exceptions (1)<br>• To make sure that good programming practices/company policies are maintained (1)<br>• To assess the efficiency of the code (1)<br>• To improve code quality (1)<br>• To share good practice/new techniques (1) | | **2** |

Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

Answer 8 of 8 · Question 4 part a.iii · 2 marks

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<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(b)</td><td>Award one mark for each correct row:
<table>
<tr><th></th><th>8-bit binary pattern</th><th>Denary value</th></tr>
<tr><td>Lowest value</td><td>1111 1111</td><td>-127</td></tr>
<tr><td>Highest value</td><td>0111 1111</td><td>(+)127</td></tr>
</table></td><td></td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(i)</td><td>Award one mark for each correct column:
<table>
<tr><th>O</th><th>S</th><th>P</th><th>O AND S</th><th>NOT P</th><th>(O AND S) AND (NOT P)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
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<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table></td><td></td><td>3</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(ii)</td><td>Award up to two marks for a linked explanation such as:<br>• ROM is non-volatile/permanent (1) therefore it retains its contents when the power is switched off / data won't be lost when power is lost (1)<br>• The contents of ROM cannot be changed (1) which makes it suitable for storing the manufacturer's instructions (1)</td><td>Explanation can be a combination of different marking points. E.g. MP1a and MP2b<br>Do not accept MP1a and MP2a or MP1b and MP2b</td><td>2</td></tr>
<tr><td colspan="3">Total for question 4</td><td>13</td></tr>
</table>

# Type MS · Finding and fixing weaknesses: penetration testing, ethical hacking, code reviews, audit trails

## Answer 8 of 8 · Question 4 part a.iii · 2 marks

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| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|